EXERCISE 1

Linear Independence of Canonical Solution Pairs via Wronskian

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Wronskian Determinant • Fundamental Bases

PHASE 1: STATEMENT & SETUP

> By finding the Wronskian of the following pairs of functions, show that they are linearly independent:
>
> 1. $x_1(t) = e^{k_1 t}$ and $x_2(t) = e^{k_2 t}$ with $k_1 \neq k_2$
> 2. $x_1(t) = e^{kt}$ and $x_2(t) = te^{kt}$
> 3. $x_1(t) = e^{\rho t}\sin(\omega t)$ and $x_2(t) = e^{\rho t}\cos(\omega t)$ with $\omega \neq 0$

PHASE 2: FUNDAMENTAL FORMULATION

For two continuously differentiable functions $x_1, x_2 \in C^1(I)$, the Wronskian determinant is defined as:

$$W[x_1, x_2](t) = \det \begin{pmatrix} x_1(t) & x_2(t) \\ \dot{x}_1(t) & \dot{x}_2(t) \end{pmatrix} = x_1(t)\dot{x}_2(t) - x_2(t)\dot{x}_1(t)$$

If $W[x_1, x_2](t_0) \neq 0$ at even a single point $t_0 \in I$, the functions $x_1(t)$ and $x_2(t)$ are strictly **linearly independent** on the interval I.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

Part (i): Distinct Exponential Modes $x_1(t) = e^{k_1 t},\ x_2(t) = e^{k_2 t}$ **(**$k_1 \neq k_2$**)**

Differentiating each function:

$$\dot{x}_1(t) = k_1 e^{k_1 t}, \qquad \dot{x}_2(t) = k_2 e^{k_2 t}$$

Formulating the Wronskian determinant:

$$W[x_1, x_2](t) = e^{k_1 t}\left(k_2 e^{k_2 t}\right) - e^{k_2 t}\left(k_1 e^{k_1 t}\right) = (k_2 - k_1)e^{(k_1+k_2)t}$$

Since the exponential function satisfies $e^{(k_1+k_2)t} > 0$ for all $t \in \mathbb{R}$, and $k_1 \neq k_2 \implies k_2 - k_1 \neq 0$:

$$W[x_1, x_2](t) \neq 0 \quad \forall t \in \mathbb{R} \implies \text{Linearly Independent on } \mathbb{R}$$

Part (ii): Repeated Root Secular Mode $x_1(t) = e^{kt},\ x_2(t) = te^{kt}$

Differentiating using the product rule on $x_2(t)$:

$$\dot{x}_1(t) = ke^{kt}, \qquad \dot{x}_2(t) = 1 \cdot e^{kt} + t(ke^{kt}) = (1 + kt)e^{kt}$$

Formulating the Wronskian determinant:

$$W[x_1, x_2](t) = e^{kt}\left[(1+kt)e^{kt}\right] - te^{kt}\left[ke^{kt}\right] = e^{2kt}\left[(1+kt) - kt\right] = e^{2kt}$$

Since $e^{2kt} > 0$ for every real t, $W(t)$ is strictly positive everywhere:

$$W[x_1, x_2](t) = e^{2kt} \neq 0 \quad \forall t \in \mathbb{R} \implies \text{Linearly Independent on } \mathbb{R}$$

Part (iii): Conjugate Complex Modes $x_1(t) = e^{\rho t}\sin(\omega t),\ x_2(t) = e^{\rho t}\cos(\omega t)$ **(**$\omega \neq 0$**)**

Differentiating via product rule:

$$\dot{x}_1(t) = e^{\rho t}\left[\rho \sin(\omega t) + \omega \cos(\omega t)\right]$$

$$\dot{x}_2(t) = e^{\rho t}\left[\rho \cos(\omega t) - \omega \sin(\omega t)\right]$$

Assembling $W(t) = x_1\dot{x}_2 - x_2\dot{x}_1$ and factoring $e^{2\rho t}$:

$$W(t) = e^{2\rho t}\left[\sin(\omega t)\left(\rho\cos(\omega t) - \omega\sin(\omega t)\right) - \cos(\omega t)\left(\rho\sin(\omega t) + \omega\cos(\omega t)\right)\right]$$

$$W(t) = e^{2\rho t}\left[\rho\sin\omega t\cos\omega t - \omega\sin^2\omega t - \rho\cos\omega t\sin\omega t - \omega\cos^2\omega t\right]$$

The cross terms $\rho\sin\omega t\cos\omega t$ cancel identically. Applying $\sin^2\omega t + \cos^2\omega t = 1$:

$$W[x_1, x_2](t) = e^{2\rho t}\left[-\omega(\sin^2\omega t + \cos^2\omega t)\right] = -\omega e^{2\rho t}$$

Because $\omega \neq 0$ and $e^{2\rho t} > 0$:

$$W[x_1, x_2](t) = -\omega e^{2\rho t} \neq 0 \quad \forall t \in \mathbb{R} \implies \text{Linearly Independent on } \mathbb{R}$$

PHASE 4: INTERPRETATION & MASTER RESULTS

MASTER WRONSKIAN SUMMARY

(i) Distinct Exponentials: $W[e^{k_1 t}, e^{k_2 t}](t) = (\mathbf{k_2} - \mathbf{k_1})\mathbf{e^{(k_1+k_2)t}} \neq 0$

(ii) Critical Damping Secular Mode: $W[e^{kt}, te^{kt}](t) = \mathbf{e^{2kt}} > 0$

(iii) Damped Harmonic Oscillations: $W[e^{\rho t}\sin\omega t, e^{\rho t}\cos\omega t](t) = -\omega\mathbf{e^{2\rho t}} \neq 0$

AEROSPACE ENGINEERING CONTEXT

These three fundamental pairs represent the complete basis spanning the null space of any linear time-invariant (LTI) mechanical system $m\ddot{x} + c\dot{x} + kx = 0$: overdamped aerodynamic divergence/decay, critically damped actuation, and flutter oscillation.